Scrum 101

🧩 What is Scrum?

Scrum is an **agile framework** used to manage and deliver projects — especially in software development.

It focuses on:

- **Working in short cycles (Sprints)**
- **Delivering small usable pieces of the product frequently**
- **Constant feedback and adaptation**

You don't plan everything at once; instead, you build a bit, learn, improve, and repeat.

Think of it like this:

"Build something small → show it → get feedback → improve → repeat."

👥 The Scrum Team

Scrum has **three key roles**:

1 Product Owner (PO) – represents the customer or business.

 * Defines *what* needs to be built.

 * Prioritizes the features (backlog items).

 * Ensures the team builds the most valuable things first.

2 Scrum Master (SM) – acts as a facilitator and coach.

 * Helps the team follow Scrum principles.

 * Removes blockers.

 * Makes sure meetings (events) happen and are effective.

3 Development Team (Dev Team) – builds the product.

 * Self-organizing and cross-functional.

 * Decides *how* to implement the features.

⚙️ The Scrum Artifacts

Artifacts are the **things you create and maintain** in Scrum — they show the progress and the plan.

1. Product Backlog

- A **list of everything** that might be needed in the product (features, bugs, enhancements).

- Owned and prioritized by the **Product Owner**.
- Continuously updated — it evolves as new ideas and feedback come in.

🧠 Example:

```
1. User can log in using email/password.
2. Add search feature.
3. Enable dark mode.
4. Improve loading speed.
```

2. Sprint Backlog

- A **subset** of the Product Backlog that the team commits to delivering in the current Sprint.
- Includes:
  - Selected user stories/features.
  - A plan (tasks) to complete them.

👷 Example:

```
Sprint 1 backlog:
1. Login screen UI
2. Backend login API
3. Integration & testing
```

3. Increment

- The **working product** delivered at the end of a Sprint.
- Must be **potentially shippable** (usable and tested).
- Combines all the completed backlog items from that Sprint.

🧱 Think of it as:

Each Sprint adds one more "brick" to the final product.

🕒 The Scrum Events (Ceremonies)

These are **time-boxed meetings** that structure the process.

1. Sprint

- A **time-boxed iteration** — typically **2 to 4 weeks**.
- Goal: deliver a **usable increment** of the product.
- Once started, the Sprint length doesn't change.

🗓️ Each Sprint follows this rhythm:

1 Plan what to build.

2 Build it.

3 Review it.

4 Reflect and improve.

2. Sprint Planning

- Held at the **start of each Sprint**.
- The **Product Owner** explains the highest-priority backlog items.
- The **team** discusses and decides **what they can commit to**.
- Output: **Sprint Goal** and **Sprint Backlog**.

🧭 **Example Sprint Goal:**

"Allow users to log in and view their profile."

3. Daily Scrum (Daily Stand-up)

- A **15-minute meeting every day** for the development team.
- Each member answers 3 questions:

  1 What did I do yesterday?

  2 What will I do today?

  3 Any blockers?

🎯 **Purpose:**
Stay aligned, detect issues early, and plan the next 24 hours.

4. Sprint Review

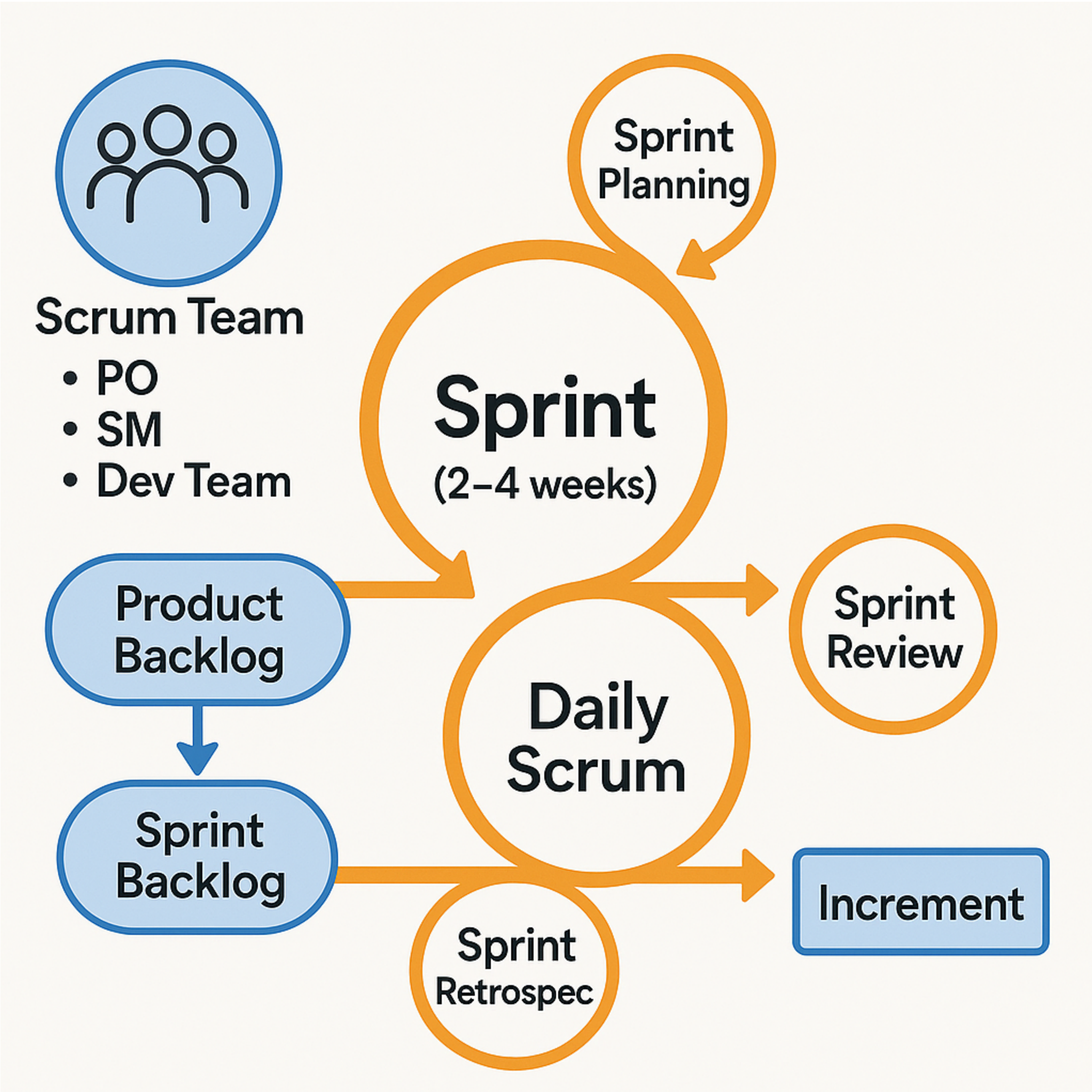
- Held **at the end of the Sprint**.
- The team **demonstrates** what was built to stakeholders.
- Get feedback → update Product Backlog.

🎬 **Think of it like a mini "product showcase."**

5. Sprint Retrospective

- Held **after the Sprint Review**, before the next Sprint Planning.
- The team reflects on the process:
  - What went well?
  - What didn't?
  - What can we improve?

💡 **Goal: Continuous improvement.**



Other Scrum Artifacts

The **core Scrum Guide** defines only *three official artifacts* (Product Backlog, Sprint Backlog, and Increment), **real-world Scrum projects** use **additional practical artifacts and concepts** like **Epics**, **User Stories**, **Story Points**, **Tasks**, etc.

🧩 Official Scrum Artifacts (Defined in Scrum Guide)

1. Product Backlog

- The master list of everything the team might ever build.
- Managed by the **Product Owner**.
- Contains **Epics**, **User Stories**, **Bugs**, **Enhancements**, etc.
- Continuously updated (called "Backlog Refinement").

2. Sprint Backlog

- The **subset** of the Product Backlog chosen for the current Sprint.
- Created in **Sprint Planning**.
- Owned by the **Development Team**.
- Contains:
    - Selected User Stories.
    - Associated Tasks (the "how" part).

3. Increment

- The **working product** (potentially shippable) that comes out of each Sprint.
- Combines all completed Product Backlog items.
- Must meet the **Definition of Done (DoD)** — a quality checklist.

⚙️ Common Practical Scrum Artifacts (Used in Real Projects)

4. Epic

- A **large feature or theme** that cannot be completed in one Sprint.
- It's **too big** to be a user story — so it's broken down later.

🧠 Example:

Epic: “User Authentication System”

This Epic can be broken into smaller **User Stories**:

- As a user, I can sign up with email.
- As a user, I can reset my password.
- As an admin, I can manage user accounts.

5. User Story

- The **building block** of the Product Backlog.
- Represents a single, small piece of functionality that delivers value to the user.
- Written in a simple, natural format.

📜 **Format:**

As a <`type of user`>**, I want** <`some goal`> **so that** <`some reason`>**.**

🧠 Example:

As a **registered user**, I want to **reset my password** so that **I can recover access to my account**.

👉 Each user story should be:

- **Independent** (can be done alone)
- **Negotiable** (details can be discussed)
- **Valuable**
- **Estimable**
- **Small**
- **Testable**

(Mnemonic: INVEST)

6. Task (or Sub-task)

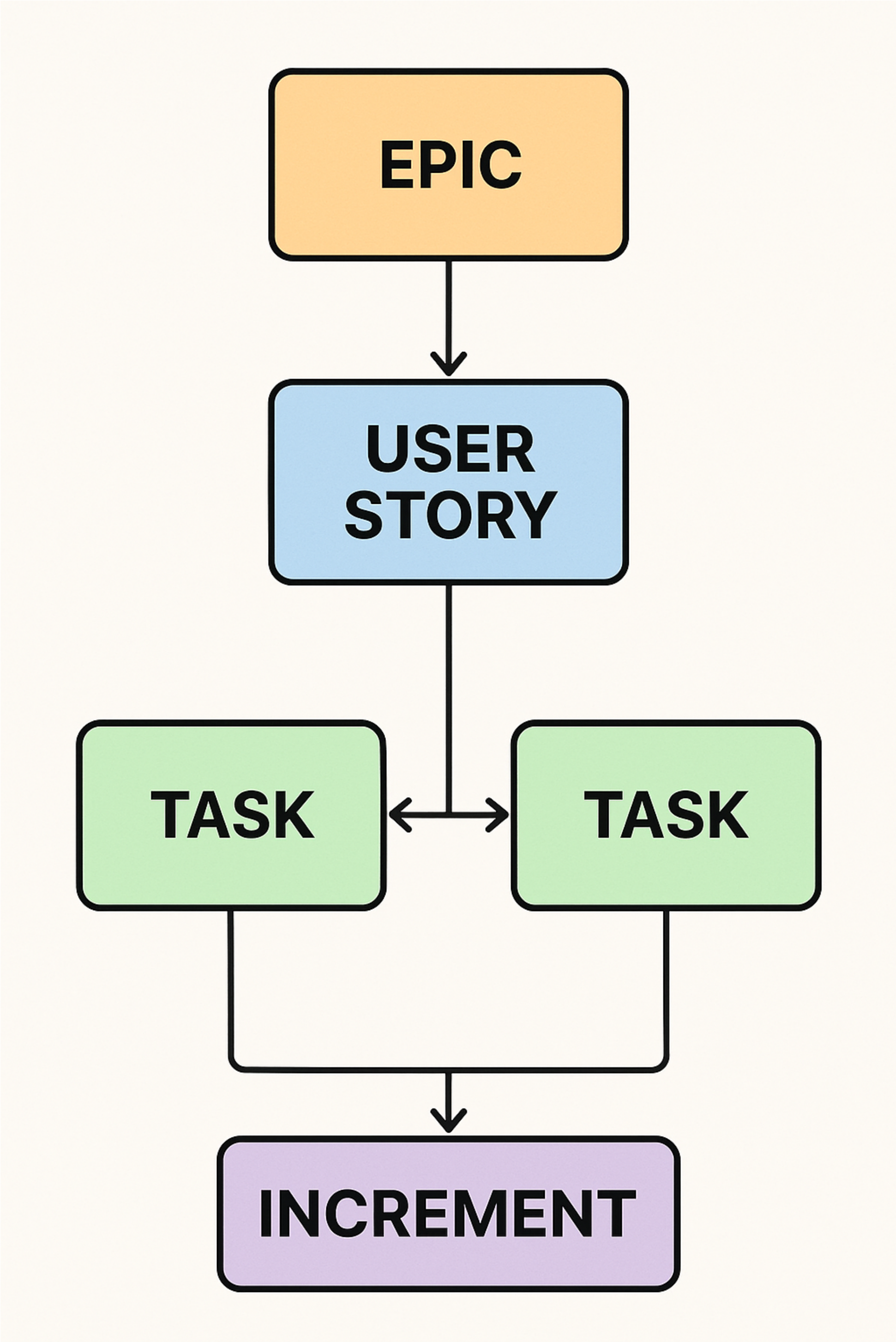
- A **smaller technical activity** that helps complete a User Story.
- Usually created during Sprint Planning.
- Helps track daily progress.

🧱 **Example:**
User Story: “User can reset password.”

Tasks:

- Create password reset API.

- Build frontend UI for reset form.
- Test reset flow.



7. Story Points

- A **relative measure of effort**, complexity, and uncertainty required to complete a story.
- Not time-based (not hours or days).
- Usually based on the **Fibonacci sequence** (1, 2, 3, 5, 8, 13...).
- Helps estimate velocity and plan future sprints.

🧮 Example:

- “Login Page” → 3 points (moderate)
- “User Dashboard” → 8 points (complex)
- “Add Contact Form” → 2 points (easy)

📈 Over time, teams track how many points they complete per Sprint — called **Velocity**.

8. Definition of Done (DoD)

- A **shared checklist** that defines when a story is truly complete.
- Ensures consistent quality.

✅ Example:

- Code implemented.
- Peer-reviewed.
- Unit tested.
- Deployed to staging.
- Documentation updated.

Only when all items are checked → the story is “Done.”

9. Acceptance Criteria

- **Specific conditions** a User Story must meet to be accepted by the Product Owner.
- Defines “what good looks like.”

🧾 Example (for “Reset Password”):

- User receives reset link by email.
- Link expires after 1 hour.
- New password must meet security rules.

10. Sprint Goal

- A **short, clear statement** of what the team aims to achieve in the Sprint.
- Acts as the guiding focus.

🎯 Example:

“Allow users to create and manage accounts.”

11. Velocity Chart

- A **visual metric** showing how many Story Points the team completed per Sprint.
- Helps predict capacity for future sprints.

📊 Example:

Sprint	Story Points Completed
1	20
2	23
3	25

12. Burndown Chart

- A graph showing **remaining work** (story points or tasks) versus **time**.
- Helps monitor Sprint progress.

📉 Ideal line: straight downward.

📉 Actual line: how the team is performing day-to-day.

🔁 Example Flow of These Artifacts in a Scrum Project

1 Product Owner creates **Epics** → breaks them into **User Stories**.

2 All stories live in the **Product Backlog**.

3 During **Sprint Planning**, team selects stories → **Sprint Backlog**.

4 Team breaks stories into **Tasks**, estimates in **Story Points**.

5 During the Sprint:

    * Team tracks progress using **Daily Scrum**, **Burndown Chart**.

6 At end of Sprint:

* Working software (the **Increment**) is delivered.
* Team reviews and improves based on **Definition of Done** and **Acceptance Criteria**.

🧠 Summary Table

Artifact	Purpose	Owner	Example
Product Backlog	Full list of desired features	Product Owner	All features & bugs
Epic	Large, high-level feature	Product Owner	“User Authentication System”
User Story	Small user-focused feature	Product Owner / Team	“Reset password”
Task	Technical step to complete story	Dev Team	“Write API endpoint”
Story Points	Measure of effort	Dev Team	“5 points”
Sprint Backlog	Stories chosen for Sprint	Dev Team	Stories planned for next 2 weeks
Definition of Done	Quality checklist	Team	“Code reviewed, tested, documented”
Increment	Working product	Team	Ready-to-demo feature
Burndown Chart	Tracks progress	Scrum Master	Remaining work trend

📈 How to Do Project Management Using Scrum

Here's how Scrum translates to project management — without traditional Gantt charts or rigid timelines.

Step 1️⃣ — Define the Product Vision

- What problem are we solving?
- What's the high-level goal of the product?
- Done by the **Product Owner** with stakeholders.

Step 2️⃣ — Build and Prioritize the Product Backlog

- Brainstorm all possible features, tasks, and bugs.
- Prioritize based on **business value** and **effort**.

Step 3️⃣ — Plan the First Sprint

- Select top items from the backlog that can fit in one Sprint.
- Define a **Sprint Goal** (clear, achievable outcome).
- Break user stories into **tasks**.

Step 4️⃣ — Execute the Sprint

- Team works on tasks.
- Hold **Daily Scrums** to track progress.
- Scrum Master removes obstacles.

Step 5️⃣ — Review and Reflect

- At the end of Sprint:
  - **Review**: Demo working software and get feedback.
  - **Retrospect**: Identify ways to work better next time.

Step 6️⃣ — Repeat

- Take lessons from the retrospective.
- Plan the next Sprint.
- Continue until product goals are met or the project ends.

📊 Tools Commonly Used for Scrum Project Management

Purpose	Tools
Backlog & Sprint management	Jira, Trello, ClickUp, Azure DevOps
Daily communication	Slack, Microsoft Teams
Documentation	Confluence, Notion
Version control	GitHub, GitLab

✅ Summary

Aspect	Description
Framework Type	Agile
Iteration	Sprint (2–4 weeks)
Roles	Product Owner, Scrum Master, Development Team
Artifacts	Product Backlog, Sprint Backlog, Increment
Events	Sprint, Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective
Goal	Deliver working software frequently and improve continuously